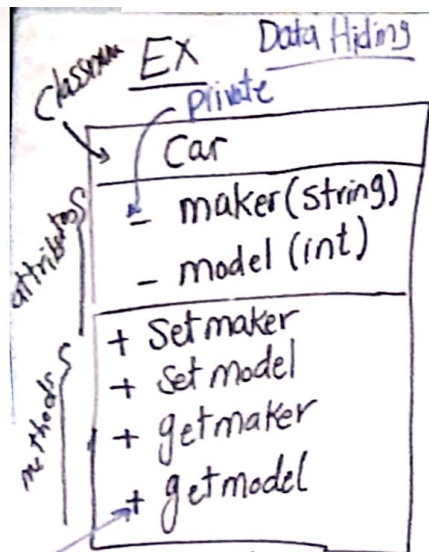




**Public UML**

int → 0  
double → 0.0  
String → null  
boolean → false

```

Public class Car {
    private String maker;
    private int model;

    Public void setmaker(String m) {
        maker = m;
    }

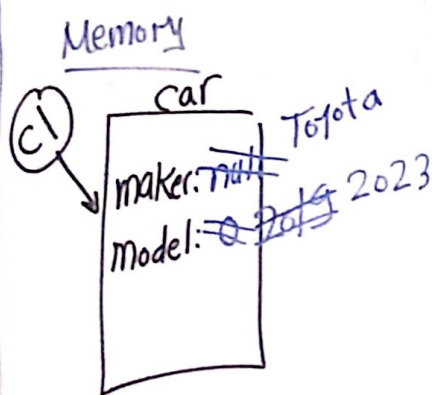
    Public void setmodel(int m) {
        model = m;
    }

    Public String getmaker() {
        return maker;
    }

    Public int getmodel() {
        return model;
    }
}

Public class test {
    Public static void main(String[] args) {
        Car c1 = new Car();
        c1.setmaker("Toyota");
        c1.setmodel(2023);
        Sy (c1.getmaker());
        Sy (c1.getmodel());
    }
}
  
```

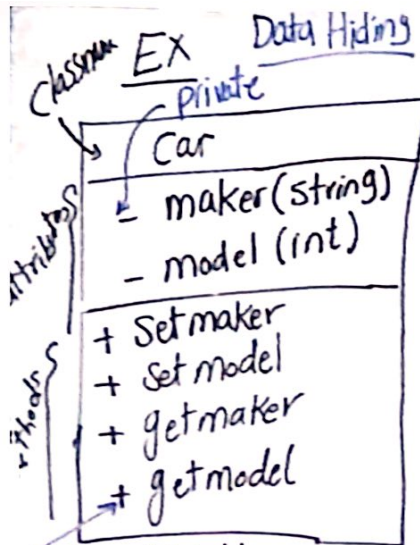
Handwritten annotations in the code above indicate the state of variables: 'maker' is set to 'Toyota' and 'model' is set to '2023' in the Car class. In the test class, 'c1' is an instance of Car, and 'Sy' represents the system output.



**output**

```

Toyota
2023
  
```



public UML

int → 0  
 double → 0.0  
 String → null  
 boolean → false

```

public class Car {
    private String maker;
    private int model;

    public void Setmaker(String m) {
        maker = m;
    }

    public void Setmodel(int m) {
        if (m > 2021) {
            else model = m;
            sy ("invalid model");
        }
    }

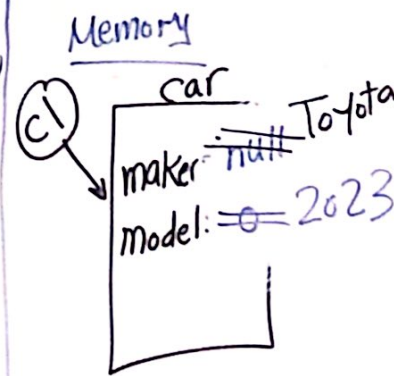
    public String getmaker() {
        return maker;
    }

    public int getmodel() {
        return model;
    }
  }
  
```

```

public class test {
    public static void main(String[] args) {
        Car c1 = new Car();
        c1.Setmaker("Toyota");
        c1.Setmodel(2019);
        c1.Setmodel(2023);
        sy (c1.getmaker());
        sy (c1.getmodel());
    }
  }
  
```

Car c1  
 c2  
 c3  
 c4



output

```

Invalid model
Toyota
2023
  
```

```

if (m == "Kia" || m == "lada" || m == "--")
    maker = m;
else
    sy ("invalid");
  
```

## Constructors:

### Special methods

- have the name as class name
- public
- No return type or void
- No return value
- Cannot be called
- Called automatic when you create object from class

### Syntax:

```
Public classname ( )  
{  
    //  
    //  
}
```

```
public class Rectangle  
{  
    private int length;  
    private int width;  
    public Rectangle ( )  
    {  
        length = 100;  
        width = 300;  
        sy — ("Hello");  
    }  
    public Rectangle (int L, int W)  
    {  
        length = L;  
        width = W;  
    }  
}
```

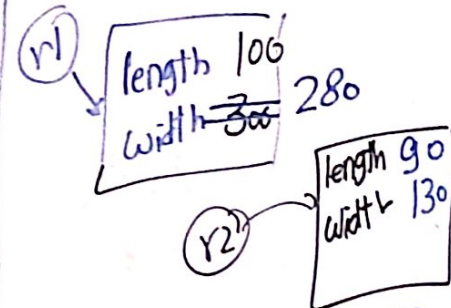
```
public void setlength (int L)  
{  
    length = L;  
}  
public void setwidth (int W)  
{  
    width = W;  
}
```

```
public int getlength ( )  
{  
    return length;  
}  
public int getwidth ( )  
{  
    return width;  
}
```

## Public class test

```
{  
    public static void main (String[] args)  
    {  
        Rectangle r1 = new Rectangle ( );  
        sy — (r1.getlength ());  
        sy — (r1.getwidth ());  
        r1.setwidth (280);  
        sy — (r1.getlength ());  
        sy — (r1.getwidth ());  
        Rectangle r2 = new Rectangle (90, 130);  
    }  
}
```

## Memory



## output

```
Hello  
100  
300  
100  
280
```

## Constructor overloading:

```
public Rectangle ( )  
{  
    length = 100;  
    width = 300;  
}
```

```
public Rectangle (int L, int W)  
{  
    length = L;  
    width = W;  
}
```

```
Rectangle r1 = new Rectangle ( );  
Rectangle r2 = new Rectangle (80, 130);  
Rectangle r3 = new Rectangle
```

| Rectangle      |
|----------------|
| - length (int) |
| - width (int)  |

+ setlength  
+ setwidth  
+ getlength  
+ getwidth

Public class test

```

    {
    Public Static void main(String[] args)
    {
    Rectangle r1 = new Rectangle();
    }
    }
  
```

8y — (r1.getLength());

130  
L, int w) Sy — (r1.getWidth());

r1.setWidth(280);

8y — (r1.getLength());

8y — (r1.getWidth());

it L) → Rectangle r2 = new Rectangle(90, 130);

it w) Rec

~~= new~~

Memory

(r1)

length 100  
width ~~300~~ 280

(r2)

length 90  
width 130

default Constructor

int → 0  
double → 0.0  
String → null  
boolean → false

output

Hello  
100  
300  
100  
280

Constructor overloading:

Public Rectangle ( )

? length = 100;  
width = 300;

Public Rectangle (int L, int w)

Rectangle

- length (int)
- width (int)